Exercise for chapter 9

Ke Hua (花 珂)*

State Key Laboratory of Solidification Processing
Northwestern Polytechnical University,
Xi'an, Shaanxi, P.R. China

*ke.hua@nwpu.edu.cn

http://teacher.nwpu.edu.cn/kehua

Chapter 9

9.1. One mole of solid Cr_2O_3 at 2500 K is dissolved in a large volume of a liquid Raoultian solution of Al_2O_3 and Cr_2O_3, in which $X_{Cr_2O_3} = 0.2$.and which is also at 2500 K. Calculate the changes in enthalpy and entropy caused by the addition. The normal melting temperature of Cr_2O_3 is 2538 K, and it can be assumed that $\Delta S_{m,Cr_2O_3} = \Delta S_{m,Al_2O_3}$.

Solution to 9.1

$$\Delta S_{\text{melting}} = \frac{\Delta H_{\text{melting}}}{T_m} \qquad \Delta H_{m,Al_2O_3} = 107500 J$$

Table A.5 Molar Heats of Melting and Transformation

Substance	Trans.	ΔH_{trans}, J	T_{trans}, K
Ag	$s \to l$	11,090	1,234
Al	$s \to l$	10,700	934
Al_2O_3	$s \to l$	107,500	2,324
Au	$s \to l$	12,600	1,338
Ba	$\alpha \to \beta$	630	648
Ba	$\beta \to l$	7,650	1,003
Cu	$s \to l$	12,970	1,356
Ca	$\alpha \to \beta$	900	716
CaF_2	$s \to l$	31,200	1,691

$$\Delta S_{m,Al_2O_3} = \frac{107500}{2324} = 46.25\, J/K = \Delta S_{m,Cr_2O_3}$$

So $$\Delta H_{m,Cr_2O_3} = T_m\, \Delta S_m = 2538 \times 46.25 = 117400\, J$$

$$Cr(s) \longrightarrow Cr(l) \qquad \Delta H = \Delta H_m = 117400 J$$

$$Cr(l) \longrightarrow Cr(in\ solution) \qquad \Delta H = \Delta \bar{H}^{Mid} = 0$$

So $$\Delta H_{Total} = 117400\, J$$

Solution to 9.1

$$Cr(s) \longrightarrow Cr(l) \qquad \Delta S = \Delta S_m = 46.25\ J/K$$

$$Cr(l) \longrightarrow Cr(in\ solution) \qquad \Delta S = \Delta S^{Mid} = -R\ln X_{Cr} = 13.38\ J/K$$

So $\quad \Delta S_{Total} = 46.25 + 13.38 = 59.63\ J/K$

9.2 When 1 mole of argon gas is bubbled through a large volume of an Fe–Mn melt of $X_{Mn} = 0.5$ at 1863 K, the evaporation of Mn into the Ar causes the mass of the melt to decrease by 1.5 g. The gas leaves the melt at a pressure of 1 atm.Calculate the activity coefficient of Mn in the liquid alloy.

Solution to 9.2

Table A.4 The Saturated Vapor Pressures of Various Substances

$\boxed{\ln p\,(\text{atm}) = -A/T + B \ln T + C}$

Substance	A	B	C	Range, K
$CaF_{2(\alpha)}$	54,350	−4.525	56.57	298–1430
$CaF_{2(\beta)}$	53,780	−4.525	56.08	1430–1691(T_m)
$CaF_{2(l)}$	50,200	−4.525	53.96	1691–2783 (T_b)
$Fe_{(l)}$	45,390	−1.27	23.93	1809 (T_m)–3330 (T_b)
$Hg_{(l)}$	7,611	−0.795	17.168	298–630 (T_b)
$Mn_{(l)}$	33,440	−3.02	37.68	1517 (T_m)–2348 (T_b)
$SiCl_{4(l)}$	3,620	—	10.96	273–333 (T_b)
$Zn_{(l)}$	15,250	−1.255	21.79	693 (T_m)–1177 (T_b)

The saturated vapor pressure of liquid M_n at 1863K is

$$P^{\ominus}_{Mn} = exp\left(\frac{-33440}{1863} - 3.02 \ln 1863 + 37.68\right) = 0.0493atm$$

The number of mole of M_n removed is $$\frac{1.50}{54.94} = 0.0273$$

The partial pressure of M_n in the gas mixture is

$$P_{Mn} = \frac{n_{Mn}}{n_{Mn} + n_{Ar}} = \frac{0.0273}{1.0273} = 0.0266$$

Solution to 9.2

So
$$a_{Mn} = \frac{P_{Mn}}{P_{Mn}^{\ominus}} = \frac{0.0266}{0.0493} = 0.539 \qquad a_i = \frac{p_i}{p_i^{\circ}}$$

$$\gamma_{Mn} = \frac{a_{Mn}}{X_{Mn}} = \frac{0.539}{0.5} = 1.08$$

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9.3 The variation, with composition, of G^{XS} for liquid Fe–Mn alloys at 1863 K is as follows

X_{Mn}	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
G^{XS}, J	395	703	925	1054	1100	1054	925	703	395

a. Does the system exhibit regular solution behavior?

b. Calculate $\bar{G}_{Fe}^{XS}$ and $\bar{G}_{Mn}^{XS}$ at $X_{Mn} = 0.6$.

c. Calculate ΔG^{M} at $X_{Mn} = 0.4$.

d. Calculate the partial pressures of Mn and Fe exerted by the alloy of $X_{Mn} = 0.2$.

Solution to 9.3

a) Calculation of $\alpha = \frac{G^{XS}}{X_{Fe}X_{Mn}}$ for each composition gives are average value of 4396J,therefore with respect toG^{xs},the system is regular.

b) $$\bar{G}_{Fe}^{XS} = \alpha X_{Mn}^2 = 4396 \times 0.6^2 = 1583\ J$$

$$\bar{G}_{Mn}^{XS} = \alpha X_{Fe}^2 = 4396 \times 0.4^2 = 703\ J$$

c) $$\Delta G^M = 8.3144 \times 1868(0.4 \ln 0.4 + 0.6 \ln 0.6) + 4396 \times 0.4 \times 0.6 = -9370J$$

Solution to 9.3

Table A.4 The Saturated Vapor Pressures of Various Substances

$\ln p\ (\text{atm}) = -A/T + B \ln T + C$

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d) At $X_{Mn} = 0.2$

$$\gamma_{Mn} = exp\left(\frac{4396 \times 0.8^2}{8.3144 \times 1863}\right) = 1.2$$

$$\gamma_A = \exp\left(\frac{\alpha X_B^2}{RT}\right) \quad (9.64)$$

$$\gamma_B = \exp\left(\frac{\alpha X_A^2}{RT}\right) \quad (9.65)$$

So $a_{Mn} = 1.2 \times 0.2 = 0.24$ $\qquad \gamma_i = \frac{a_i}{X_i}$

$$P^{\theta}_{Mn} = exp\left(\frac{-33440}{1863} - 3.02 \ln 1863 + 37.68\right) = 0.0493\ atm$$

So $P_{Mn} = a_{Mn} P^{\theta}_{Mn} = 0.24 \times 0.0493 = 0.0018\ atm$

Solution to 9.3

$$\gamma_{Fe} = exp\left(\frac{4396 \times 0.2^2}{8.3144 \times 1863}\right) = 1.011$$

So $$a_{Fe} = 1.011 \times 0.8 = 0.809$$

$$P_{Fe}^{\theta} = exp\left(\frac{-45390}{1863} - 1.27 \ln 1863 + 23.93\right) = 4.55 \times 10^{-5}\ atm$$

So $$P_{Fe} = a_{Fe}P_{Fe}^{\theta} = 0.809 \times 4.55 \times 10^{-5} = 3.68 \times 10^{-5}\ atm$$

9.4 Calculate the heat required to form a liquid solution at 1356 K, starting with 1 mole of Cu and 1 mole of Ag at 298 K. At 1356 K, the molar heat of mixing of liquid Cu and liquid Ag is given by $\Delta H^{M} = -20590 X_{Cu} X_{Ag}$

Solution to 9.4

Heat 1 mole of Cu from 298K to 1356K and melt it

$$\Delta H = \int_{298}^{1356} C_{p,Cu}\, dT + \Delta H_{m,Cu}$$

$$= 22.64(1356 - 298) + \frac{6.28 \times 10^{-3}}{2}(1356^2 - 298^2) + 12970$$

$$= 42420J$$

Solution to 9.4

Heat 1 mole of Ag from 298K to 1356K and melt it

$$\Delta H = \int_{298}^{1234} C_{p,\mathrm{Ag}(s)}\, dT + \int_{1234}^{1356} C_{p,\mathrm{Ag}(l)}\, dT + \Delta H_{m,Ag}$$

$$= 21.3 \times (1234 - 298) + \frac{8.54 \times 10^{-3}}{2}(1234^2 - 298^2)$$

$$- 1.51 \times 10^5 \left(\frac{1}{1234} - \frac{1}{298}\right) + 30.5(1356 - 1234) + 11090 = 41260J$$

In 2 moles of equimolar liquid solution $\Delta H^M = -20590 X_{Cu} X_{Ag}$

$$\Delta H = 2(-20590 \times 0.5^2) = -10360J$$

So total requirement $= 42420 + 41260 - 10360 = 73380J$

9.5 Melts in the system Pb–Sn exhibit regular solution behavior. At 473°C, $a_{Pb} = 0.055$ in a liquid solution of $X_{Pb} = 0.1$. Calculate the value of α for the system and calculate the activity of Sn in the liquid solution of $X_{Sn} = 0.5$ at 500°C .

Solution to 9.5

$$\gamma_{Pb} = \frac{a_{Pb}}{X_{Pb}} = \frac{0.055}{0.1} = 0.55$$

$$RT \ln \gamma_A = \Delta \overline{H}_A^M = \alpha X_B^2$$

So $$\alpha = \frac{8.3144 \times 746 \times \ln 0.55}{0.9^2} = -4578\ J$$

At 500℃ (773K) and $X_{Sn} = 0.5$

$$RT \ln \gamma_A = \Delta \overline{H}_A^M = \alpha X_B^2$$

$$\ln \gamma_{Sn} = \frac{-4578 \times 0.5^2}{8.3144 \times 773} = -0.178$$

So $$a_{Sn} = 0.5 \times \exp(-0.178) = 0.418$$

$$\gamma_i = \frac{a_i}{X_i}$$

9.6

a. Calculate the values of $\Delta\bar{G}_B$ and a_B for an alloy of $X_B = 0.5$ at 1000 K assuming the solution is ideal .

b. Calculate the values of $\Delta\bar{G}_B$ and a_B for an alloy of $X_B = 0.5$ at 1000 K that is a regular solution with $\Delta H_{mixing} = 16628 X_A X_B$.

Solution to 9.6

a) For ideal and regular solutions $\Delta G^M\ (X_B = 0.5) = \Delta \bar{G}_A^M = \Delta \bar{G}_B^M$

$$\Delta G^{M,\mathrm{id}} = RT(X_A \ln X_A + X_B \ln X_B)$$

$$\Delta G^{M,id}(X_B = 0.5) = -RT\Delta S^M\ (X_B = 0.5)$$

$$= -8314\left(\frac{1}{2}\ln\frac{1}{2} + \frac{1}{2}\ln\frac{1}{2}\right) = -5763\ J$$

$$\Delta \bar{G}_B = RT\ln a_B = -5763\ J$$

$$a_B(X_B = 0.5) = exp\left(-\frac{5763}{8314}\right) = 0.5$$

Solution to 9.6

$$\Delta G^M = \alpha X_A X_B - T\Delta S^{M,\text{ideal}} \qquad \Delta S^M = -R\left(X_A \ln X_A + X_B \ln X_B\right)$$

b)

$$\Delta G^{M,reg}(0.5) = 16626 \times \frac{1}{4} + 8314\left(\frac{1}{2}\ln\frac{1}{2} + \frac{1}{2}\ln\frac{1}{2}\right)$$

$$= -1609\ J$$

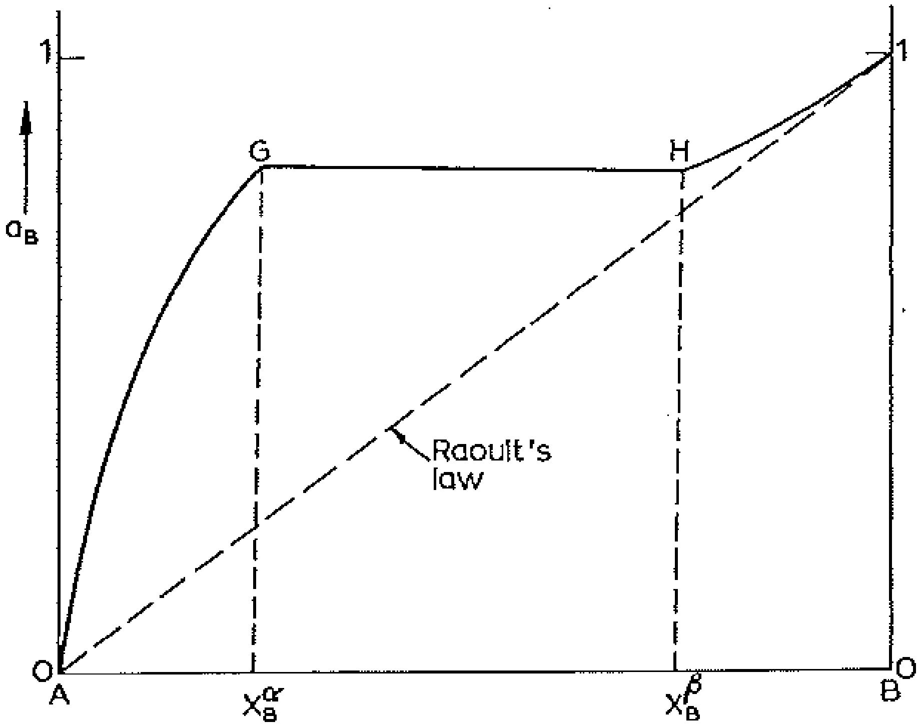
$$\Delta \bar{G}_B = RT\ln a_B = -1609\ J$$

$$a_B(X_B = 0.5) = exp\left(-\frac{1609}{8314}\right) = 0.824$$

9.7 A regular solution exhibits a miscibility gap. Sketch the activity of X_B vs. X_B at a temperature within the miscibility gap.

Denote regions where Henry's law is obeyed and where Raoults' law is obeyed if applicable .

Solution to 9.7



$\longrightarrow X_B$

$\longleftarrow X_A$

9.8 Tin obeys Henry's law in dilute liquid solutions of Sn and Cd, and the Henrian activity coefficient of Sn, γ°_{Sn} , varies with temperature as

$$\ln \gamma^{\circ}_{Sn} = -\frac{840}{T} + 1.58$$

Calculate the change in temperature when 1 mole of liquid Sn and 99 moles of liquid Cd are mixed in an adiabatic enclosure. The molar constant-pressure heat capacity of the alloy formed is 29.5 J/K

Solution to 9.8

$$\frac{\partial(R\ln\gamma_i)}{\partial\left(\frac{1}{T}\right)} = \Delta\overline{H}_i^M$$

$$\Delta\overline{H}_{Sn}^M = \frac{R\partial\ln\gamma_{Sn}^{\circ}}{\partial\left(\frac{1}{T}\right)} = -8.3144 \times 840 = -6984\ J$$

$$\Delta H^M = X_A\Delta\overline{H}_A^M + X_B\Delta\overline{H}_B^M$$

$$\Delta H^M(per\ mole) = X_{Sn}\Delta\overline{H}_{Sn}^M + X_{Cd}\Delta\overline{H}_{Cd}^M \qquad \Delta\overline{H}_{Cd}^M = 0$$

$$= 0.01 \times (-6984) = -69.84\ J = C_P\Delta T$$

$$= 29.5\Delta T$$

So

$$\Delta T = \frac{69.84}{29.5} = 2.37\ K$$

9.9 Use the Gibbs–Duhem equation to show that, if the activity coefficients of the components of a binary solution can be expressed as

$$\ln\gamma_A = \alpha_1 X_B + \frac{1}{2}\alpha_2 X_B^2 + \frac{1}{3}\alpha_3 X_B^3 + \cdots$$

and

$$\ln\gamma_{\mathrm{B}} = \beta_1 X_{\mathrm{A}} + \frac{1}{2}\beta_2 X_{\mathrm{A}}^2 + \frac{1}{3}\beta_3 X_{\mathrm{A}}^3 + \cdots$$

over the entire range of composition, then $\alpha_1 = \beta_1 = 0$, and that, if the variation can be represented by the quadratic terms alone, then $\alpha_2 = \beta_2$.

Solution to 9.9

$$\ln\gamma_A = \alpha_1 X_B + \frac{1}{2}\alpha_2 X_B^2 + \frac{1}{3}\alpha_3 X_B^3 + \cdots$$

So $$X_A \mathrm{d}\ln\gamma_A = \alpha_1 X_A \mathrm{d}X_B + \alpha_2 X_A X_B \mathrm{d}X_B + \alpha_3 X_A X_B^2 \mathrm{d}X_B + \cdots$$

$$\ln\gamma_\mathrm{B} = \beta_1 X_\mathrm{A} + \frac{1}{2}\beta_2 X_\mathrm{A}^2 + \frac{1}{3}\beta_3 X_\mathrm{A}^3 + \cdots$$

So $$X_B \mathrm{d}\ln\gamma_\mathrm{B} = \beta_1 X_B \mathrm{d}X_\mathrm{A} + \beta_2 X_B X_A \mathrm{d}X_A + \beta_3 X_B X_\mathrm{A}^2 \mathrm{d}X_A + \cdots$$

For the Gibbs–Duhem equation

$$X_A \mathrm{d}\ln\gamma_A = -X_B \mathrm{d}\ln\gamma_\mathrm{B}$$

Solution to 9.9

So $$\alpha_1 X_A \mathrm{d}X_B + \alpha_2 X_A X_B \mathrm{d}X_B = \beta_1 X_B \mathrm{d}X_\mathrm{A} + \beta_2 X_B X_A \mathrm{d}X_A$$

If the power series are fluctuated at the quadratic terms.

Equaling of the coefficients in terms containing X_A (or X_B) requires that $\alpha_1 = \beta_1 = 0$ and equaling of the coefficients in the quadratic terms requires that $\alpha_2 = \beta_2$.

9.10 The activity coefficient of Zn in liquid Zn–Cd alloys at 435°C can be represented as

$$\ln \gamma_{\mathrm{Zn}} = 0.875 X_{\mathrm{Cd}}^2 - 0.30 X_{\mathrm{Cd}}^3$$

Derive the corresponding expression for the dependence of $\ln \gamma_{Cd}$ on composition and calculate the activity of cadmium in the alloy of $X_{\mathrm{Cd}} = 0.5$ at 435°C.

Solution to 9.10

$$\ln \gamma_{\mathrm{Zn}} = 0.875X_{\mathrm{Cd}}^2 - 0.30X_{\mathrm{Cd}}^3$$

$$d\ln \gamma_{Zn} = 1.750X_{Cd}dX_{Cd} - 0.9X_{Cd}^2 dX_{Cd}$$

$$-\frac{X_{Zn}}{X_{Cd}} d\ln \gamma_{Zn} = -1.75\, X_{Zn}dX_{Cd} + 0.9X_{Zn}X_{Cd}dX_{Cd}$$

$$= 1.75X_{Zn}dX_{Zn} - 0.9X_{Zn}(1 - X_{Zn})dX_{Zn}$$

$$= 0.85X_{Zn}dX_{Zn} + 0.9X_{Zn}^2 dX_{Zn}$$

$$X_A \mathrm{dln}\, \gamma_A = -X_B \mathrm{dln}\, \gamma_{\mathrm{B}}$$

So $\ln \gamma_{\mathrm{Cd}} = 0.425X_{\mathrm{Zn}}^2 + 0.30X_{\mathrm{Zn}}^3$

So At $X_{\mathrm{Cd}} = 0.5, \gamma_{\mathrm{Cd}} = 1.155$ and $a_{\mathrm{Cd}} = 0.577$

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~~9.11~~ The molar excess Gibbs free energy of formation of solid solutions in the system Au–Ni can be represented by

$$G^{XS} = X_{Ni}X_{Au}(24140X_{Au} + 38280X_{Ni} - 14230X_{Au}X_{Ni})\left(1 - \frac{T}{2660}\right) J$$

Calculate the activities of Au and Ni in the alloy of $X_{\mathrm{Au}} = 0.5$ at 1100 K.

Solution to 9.11

Let $X_{Au} = x$

$$G^{XS} = X_{Ni}X_{Au}(24140X_{Au} + 38280X_{Ni} - 14230X_{Au}X_{Ni})\left(1 - \frac{T}{2660}\right)\ J$$

$$G^{XS} = \mathrm{x}(1 - \mathrm{x})[24140x + 38280(1 - x) - 14230x(1 - x)]\left(1 - \frac{1100}{2660}\right)\ J$$

$$= [38280\mathrm{x} - 66650x^2 + 42600x^3 - 14230x^4] \times 0.586 \qquad (1)$$

$$\frac{\partial G^{XS}}{\partial x} = [38280 - 133300x + 127800x^2 - 56920x^3] \times 0.586$$

$$(1 - x)\frac{\partial G^{XS}}{\partial x} = [38280 - 171580x + 261100x^2 - 184720x^3 + 56920x^4] \times 0.586$$

Solution to 9.11

$$\bar{G}_{Au}^{XS} = G^{XS} + (1-x)\frac{\partial G^{XS}}{\partial x} \qquad \Delta\bar{G}_B^M = \Delta G^M + X_A \frac{d\Delta G^M}{dX_B}$$

$$= [38280 - 133300x + 194450x^2 - 142120x^3 + 42690x^4] \times 0.586$$

So $$\bar{G}_{Au}^{XS} = 3105\,J = 8.3144 \times 1100 \ln \gamma_{Au} \qquad \bar{G}_A^{XS} = RT\ln\gamma_A$$

$$\gamma_{Au} = 1.39$$

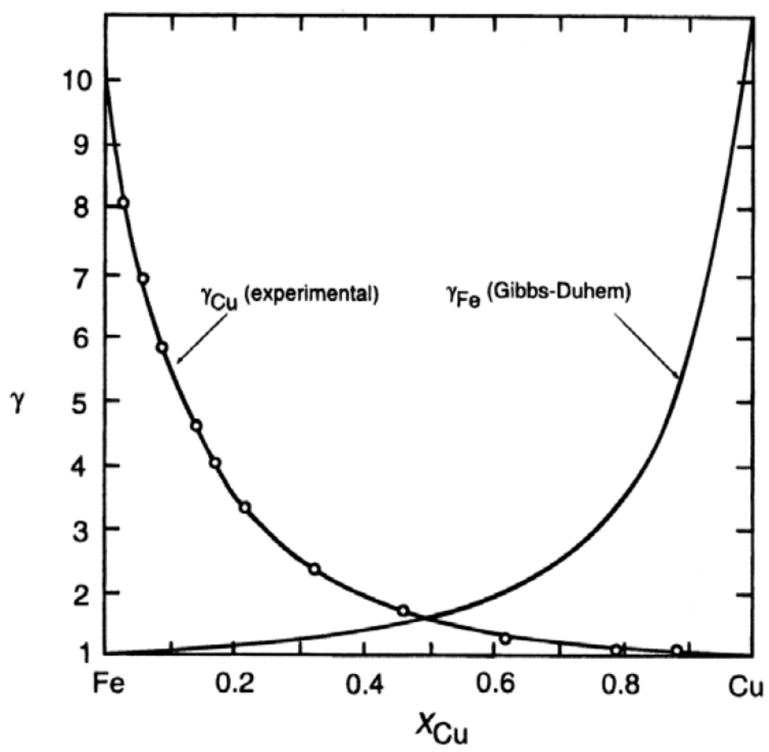
$$a_{Au} = 0.5 \times 1.39 = 0.695 \qquad \gamma_i = \frac{a_i}{X_i}$$

From (1), At $X_{\mathrm{Au}} = 0.5, G^{XS} = 4051\,\mathrm{J} = 8.3144 \times 1100 \times [0.5\ln a_{Ni} + 0.5\ln a_{Au}]$

So $$a_{Ni} = 0.85$$

9.12 The activities of liquid iron–copper solutions at 1550°C as a function of composition are shown in Figure 9.9. Sketch the activity coefficients as a function of composition at this temperature

Solution to 9.12



9.13 Use the equation $\Delta G^{XS} = b(T - T_C)\eta^2 + c\eta^4$ to derive the following equation:

For $\eta = \eta(T)$: $\eta_{eq} = \left(1 - \frac{T}{T_C}\right)^{1/2}$ Where $b > 0$

Hint: Use the Third Law of Thermodynamics to set $\eta = 1$ at $\mathrm{T} = 0$.

Solution to 9.13

$$\Delta G^{XS} = b(T - T_C)\eta^2 + c\eta^4$$

$$\frac{\partial \Delta G^{XS}}{\partial \eta} = 0$$ for equilibrium

$$\frac{\partial \Delta G^{XS}}{\partial \eta} = 2b(T - T_C)\eta + 4c\eta^3$$

Thus $\eta = 0$ and $\eta^2 = -\frac{b(T-T_C)}{2C}$ are extrema

Solution to 9.13

By taking the second derivative $\eta = 0$ can be shown to be a maximum when T< Tc and the other two extrema are minima when T< Tc.

$$\eta^2 = -\frac{b(T-T_C)}{2C}$$

At $\mathrm{T} = 0$, $\eta = 1$ (by the third law)

Thus: $T_C = \frac{2C}{b}$ which leads to $\eta^2 = \left(1 - \frac{T}{T_C}\right)$

9.14 It has been found that in a certain solution, the activity $a_A = X_A$ over a certain range of composition. Determine the relationship between a_B and X_B over the same range of composition.

Solution to 9.14

$$a_A = X_A$$

$$d(\ln a_A) = d(\ln X_A)$$

$$X_A d(\ln a_A) + X_B d(\ln X_B) = 0 \qquad \text{Gibbs-Duhem}$$

Thus $$d(\ln a_2) = -\frac{X_A}{X_B} d(\ln a_A) = -\frac{X_A}{X_B} d(\ln X_A) = \frac{dX_B}{X_B}$$

Integrating $$\ln a_2 = \ln X_2 + \ln \gamma$$ Where γ is a constant

Thus $$a_B = \gamma X_B = \gamma_B X_B$$

This is Herry's law

9.15 All regular solutions with positive heats of mixing have the same value of the activity of its components (A or B) at the critical point of the miscibility gap.

a. Calculate this activity.

Solution to 9.15

a)

$$a_B = \gamma_B X_B \qquad\qquad \gamma_A = \exp\left(\frac{\alpha X_B^2}{RT}\right) \qquad (9.64)$$

$$a_B = exp\left(\frac{a_0 X_B^2}{RT}\right) X_B$$

$$T_C = \frac{a_0}{2R}$$

$$a_B = exp\left(\frac{a_0\left(\frac{1}{2}\right)^2}{R\frac{a_0}{2R}}\right)\left(\frac{1}{2}\right) = exp(0.5)\left(\frac{1}{2}\right) = 0.824$$

9.16 At a certain temperature, T, the A–B system exhibits regular solution behavior. The activity coefficient of A is given by

$$\ln \gamma_A = -b(1 - X_A)^2$$

where b is a constant at the given T.

Compute the corresponding equation for the variation of γ_B with composition at the same temperature. Be sure to state the justification for the steps of your solution.

Solution to 9.16

$$\ln \gamma_A = -b(1 - X_A)^2$$

$$X_A d\bar{G}_A + X_B d\bar{G}_B = 0$$

$$d\bar{G}_A = RTd \ln a_A = RTd \ln(\gamma_A X_A)$$

$$X_A d \ln \gamma_A + X_B d \ln \gamma_B = 0$$

$$d \ln \gamma_B = -\frac{X_A}{X_B} d \ln \gamma_A = -\frac{X_A}{X_B}(-2b)(1 - X_A)(-dX_A)$$

$$= -2b\frac{X_A}{X_B}X_B \mathrm{d}X_A = -2bX_A \mathrm{d}X_A$$

So $$\int_{\gamma}^{\gamma_B} d \ln \gamma_B = \ln \gamma_B = -bX_A^2 = -b(1 - X_B)^2$$

The End